

Fast-track convergence of the near-unit inverse signed tower

Lemma 1 (Fast-track convergence of the near-unit inverse signed tower). *Let $c > 0$, $L = \log c$, and*

$$T_c(z) = c^{c^{-z}} = \exp(Le^{-Lz}).$$

There exist $\rho, r > 0$ such that, for every real L with $|L| < \rho$, the map T_c has a unique fixed point $z_+(c)$ in the disk $|z - 1| < r$. This fixed point is real-analytic in L , hence in c near $c = 1$, and satisfies

$$z_+(c) = 1 + L - \frac{1}{2}L^2 - \frac{4}{3}L^3 + O(L^4).$$

In particular,

$$z_+(c) = 1 + L - \frac{1}{2}L^2 + O(L^3).$$

The multiplier at the fixed point is

$$\lambda(c) = T'_c(z_+) = -L^2 e^{-Lz_+} z_+,$$

and satisfies

$$\lambda(c) = -L^2 + 2L^4 + O(L^5), \quad |\lambda(c)| = L^2(1 + O(L^2)).$$

Thus, after possibly shrinking ρ , $z_+(c)$ is attracting for $0 < |L| < \rho$, and $|\lambda(c)| \rightarrow 0$ as $c \rightarrow 1$.

If z_0 belongs to the basin of attraction of $z_+(c)$, and $z_{n+1} = T_c(z_n)$, then for $0 < |L| < \rho$ there is a constant $C = C(c, z_0)$ such that

$$|z_n - z_+(c)| \leq C|\lambda(c)|^n = C|L|^{2n}(1 + O(L^2))^n.$$

Equivalently, for a depth- N truncation,

$$|z_N - z_+(c)| \leq C|L|^{2N}(1 + O(L^2))^N.$$

Thus the convergence rate is governed by the near-unit parameter $|\log c|^2$. Lower-level tower structure can affect the prefactor through the initial seed z_0 , but not the asymptotic local rate once the orbit is in the attracting basin.

Proof. Set

$$F(z, L) = z - \exp(Le^{-Lz}).$$

Then

$$F(1, 0) = 0, \quad \partial_z F(1, 0) = 1 \neq 0.$$

By the implicit function theorem, there exist $\rho, r > 0$ and a unique analytic function $z_+(L)$, defined for $|L| < \rho$, such that $|z_+(L) - 1| < r$ and

$$F(z_+(L), L) = 0.$$

Since $L = \log c$ is analytic for c near 1, the fixed point is also analytic in c .

To compute the expansion, write

$$z_+(L) = 1 + aL + bL^2 + dL^3 + O(L^4).$$

Substituting into

$$z_+ = \exp(Le^{-Lz_+})$$

and comparing coefficients gives

$$a = 1, \quad b = -\frac{1}{2}, \quad d = -\frac{4}{3}.$$

Hence

$$z_+(L) = 1 + L - \frac{1}{2}L^2 - \frac{4}{3}L^3 + O(L^4).$$

Next,

$$T'_c(z) = \exp(Le^{-Lz}) \cdot L(-L)e^{-Lz} = -L^2e^{-Lz}T_c(z).$$

Evaluating at the fixed point, where $T_c(z_+) = z_+$, gives

$$\lambda(c) = T'_c(z_+) = -L^2e^{-Lz_+}z_+.$$

Using

$$z_+ = 1 + L + O(L^2), \quad e^{-Lz_+} = 1 - L + O(L^2),$$

we get

$$e^{-Lz_+}z_+ = 1 + O(L^2),$$

so

$$|\lambda(c)| = L^2(1 + O(L^2)).$$

A sharper expansion gives

$$e^{-Lz_+}z_+ = 1 - 2L^2 + O(L^3),$$

hence

$$\lambda(c) = -L^2 + 2L^4 + O(L^5).$$

Therefore $|\lambda(c)| < 1$ for sufficiently small nonzero L .

For $0 < |L| < \rho$, the fixed point is attracting and non-superattracting. By Koenigs linearization, there is a local analytic coordinate ϕ near z_+ , with $\phi(z_+) = 0$ and $\phi'(z_+) \neq 0$, such that

$$\phi(T_c(z)) = \lambda(c)\phi(z).$$

If z_0 is in the basin of attraction, then for some m , z_m lies in the linearization neighborhood. Hence, for $n \geq m$,

$$|\phi(z_n)| = |\lambda(c)|^{n-m}|\phi(z_m)|.$$

Since ϕ^{-1} is locally Lipschitz near 0, there is a constant K such that

$$|z_n - z_+| \leq K|\phi(z_n)| \leq K|\lambda(c)|^{n-m}|\phi(z_m)|.$$

Absorbing the finite initial segment and the factor $|\lambda|^{-m}$ into a constant $C = C(c, z_0)$, we obtain

$$|z_n - z_+(c)| \leq C|\lambda(c)|^n.$$

Using $|\lambda(c)| = |L|^2(1 + O(L^2))$ gives

$$|z_n - z_+(c)| \leq C|L|^{2n}(1 + O(L^2))^n.$$

Taking $n = N$ gives the truncation bound. □